Calculated fields, Parameters & LOD Expression

Question 1 :

What is a Calculated Field and why is it important in business dashboards?

Ans : A calculated field in Tableau is a custom field created using formulas based on existing data. It allows users to perform calculations such as mathematical operations, logical conditions, and aggregations directly within Tableau.

Calculated fields are important in business dashboards because they help transform raw data into meaningful insights. They enable the creation of key performance indicators (KPIs), ratios, and metrics that are not directly available in the dataset.

For example, a business may use a calculated field to compute Profit Ratio using Profit divided by Sales, which helps in evaluating business performance more effectively.

Overall, calculated fields enhance data analysis, improve decision-making, and allow users to customize dashboards according to business needs

Question 2 :

Why can't parameters change visuals directly in Tableau?

Ans : Parameters in Tableau cannot directly change visuals because they are independent variables and are not connected to the underlying data source. Unlike filters, parameters do not automatically interact with the data or visualization.

Parameters only store a single value selected by the user. To affect a visual, they must be used within calculated fields, filters, or reference lines. These calculations then apply the parameter value to the data, which in turn updates the visualization.

Therefore, parameters act as an input mechanism, but they require calculated fields to influence the visual output.

Question 3 :

When should EXCLUDE LOD be used?

Ans : EXCLUDE LOD is used when we want to remove certain dimensions from the level of detail in a calculation. I use it when I need higher-level aggregates, such as region totals, while still displaying more detailed dimensions like category in the view.

Question 4 :

Difference between FIXED and INCLUDE LOD expressions?

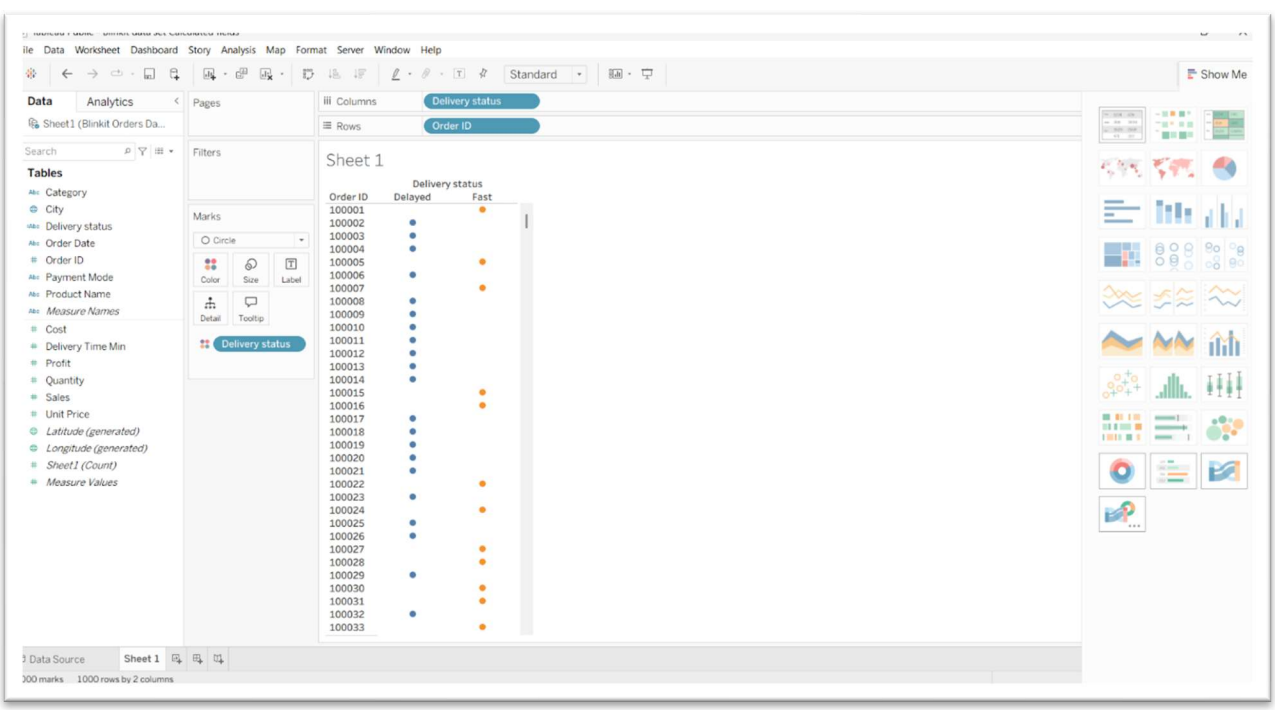
Ans : **FIXED LOD** calculates values at a specific level of detail, independent of the view. It ignores the dimensions present in the visualization.

INCLUDE LOD adds extra dimensions to the calculation along with the dimensions in the view, making it dependent on the visualization.

In short, FIXED sets a fixed level of aggregation, while INCLUDE extends the level of detail.

Task 1: Delivery Performance Classification (Calculated Field) Business Need Operations wants to track delivery efficiency. Task Classify orders as Fast or Delayed.

Condition Delivery ≤ 30 min → Fast Else → Delayed



Task 2: Loss-Making Orders Identification

Business Need High sales do not always mean high profit. Task Identify orders resulting in loss

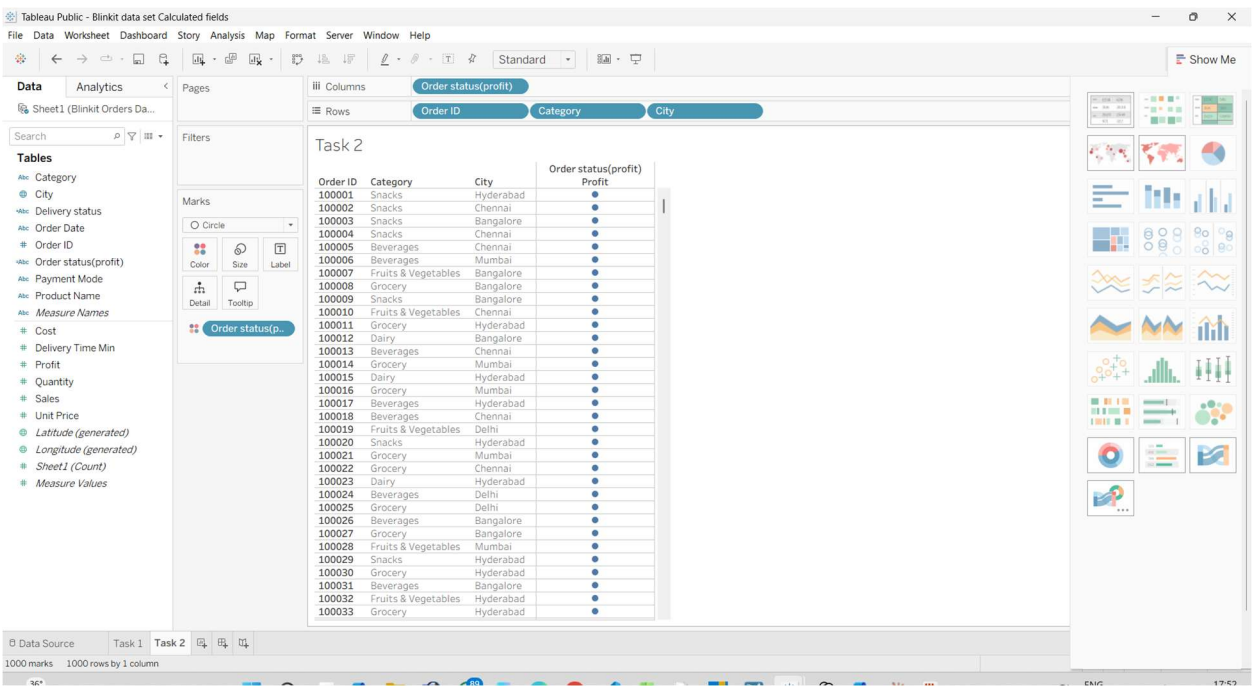
Ans : first of all, I will create a calculated field i.e.

IF [Profit] < 0 THEN “Loss”

ELSE “Profit”

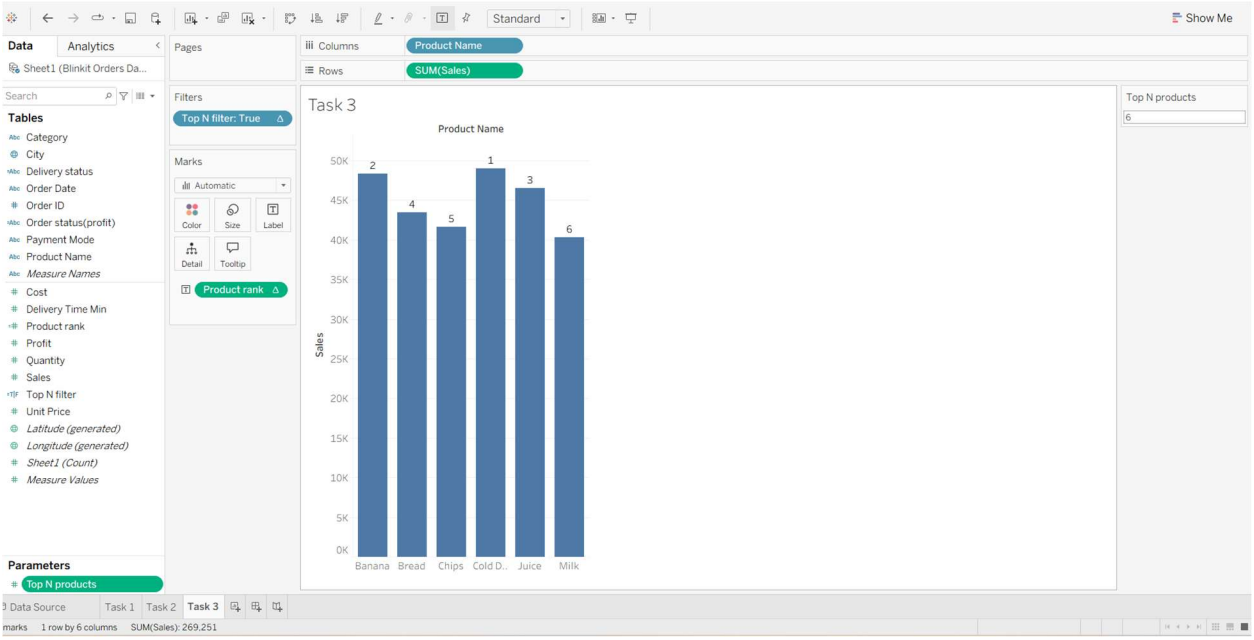
END

Then, we get



Task 3: Top N Products Analysis (Parameter) Business Need The product team wants to focus on top-performing products. Task Create a Top N Products dynamic analysis

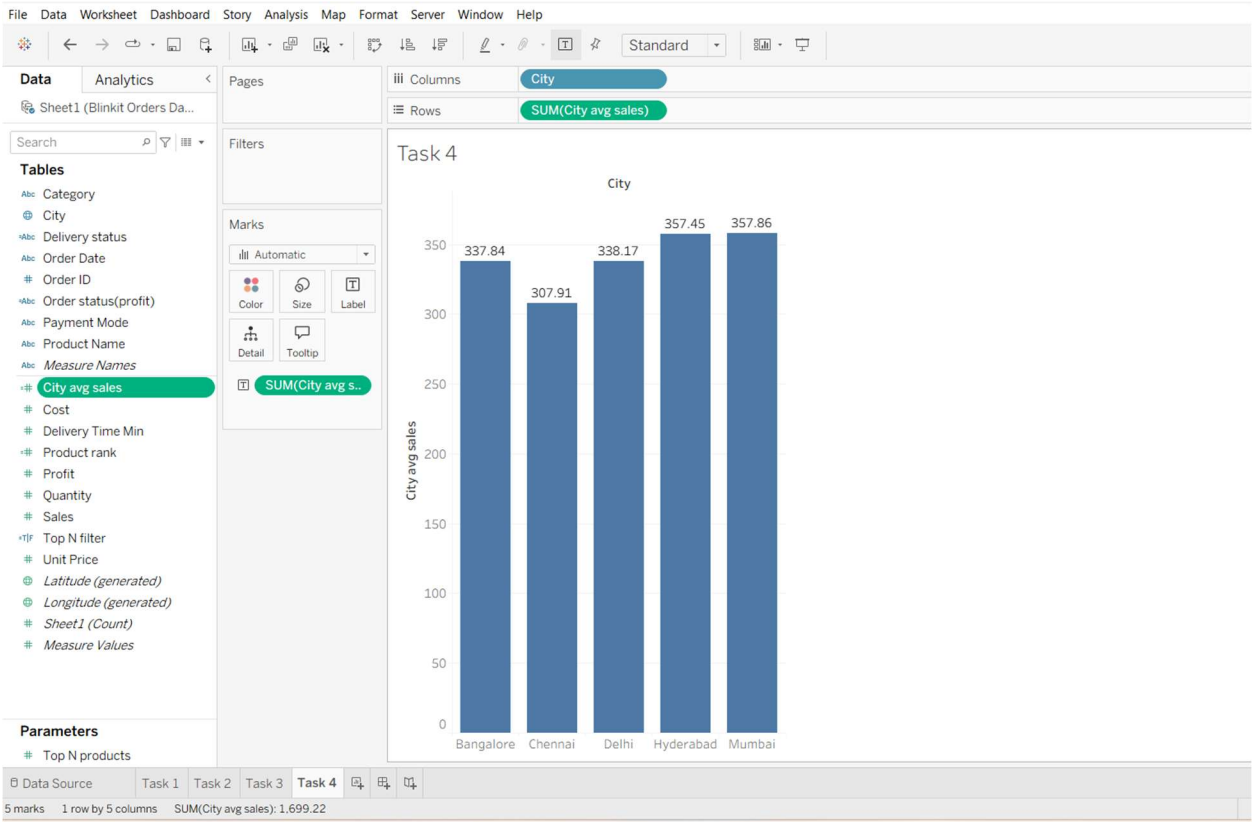
Ans :



Task 4: City-Level Benchmarking (FIXED LOD) Business Need Compare city performance independent of category filters. Task Calculate average sales per city using FIXED LOD.

Ans :

I used here this syntax : { FIXED [City] : AVG([Sales]) }


File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Standard
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Category
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Delivery status
Order Date
Order ID
Order status(profit)
Payment Mode
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Longitude (generated)
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Size
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Task 4
City
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300
250
200
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50
0
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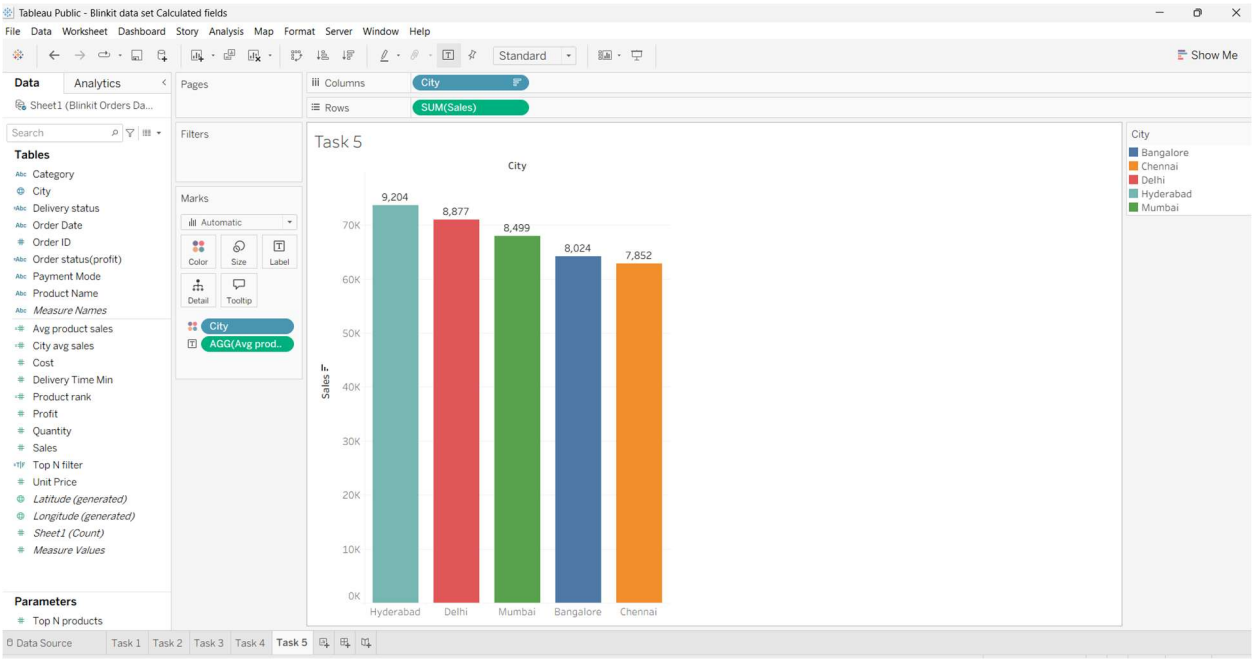
307.91
338.17
357.45
357.86
Bangalore
Chennai
Delhi
Hyderabad
Mumbai
Data Source
Task 1
Task 2
Task 3
Task 4
5 marks
1 row by 5 columns
SUM(City avg sales): 1,699.22

Task 5: Customer/Product Detail Impact (INCLUDE LOD) Business Need The sales team wants average sales considering product-level detail, even when product is not shown. Task Create a calculation that finds the average sales at the product level, but allows the result to be used at a higher level (such as Category or Region). Use INCLUDE LOD.

Ans :

Here I used this LOD expression :

Avg Product Sales (LOD)

AVG({ INCLUDE [Product Name] : SUM([Sales]) })